

# Analysis of Symmetry of 2-point Correlation Functions in Finite Temperature QCD

Using Möbius Domain-Wall Fermions

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# Outline

- Research Background
- Theoretical Background
- Lattice Simulation Setup
- Numerical Results
- Summary and Outlook

# Research Background

- The behavior of  $U(1)_A$  axial symmetry restoration is important to the order of the chiral phase transition (and universality class if it is 2nd-order), so as our prediction capability with perturbative method
- The historical effective linear sigma model analysis showed the  $U(1)_A$  can not be restored before the transition, if the transition is second order, due to the non-existence of stable IR fixed point under non vanish anomalous  $U(1)_A$  effect in 2 flavor case
- Previous Lattice simulation study, such as HotQCD, concluded that the chiral phase transition is second order and extracted the universal coefficient, while  $U(1)_A$  is restored near  $1.5T_c \sim 2T_c$

# Motivation for this work

- However, the higher loop linear sigma model study and Functional Renormalization Group analysis showed there's nontrivial stable IR fixed point and also the possibility of weak 1st order transition. So, the observation of universal behavior is not enough to conclude the  $U(1)_A$  symmetry behavior and also the universality class
- The previous study use Highly Improved Staggered fermion that breaks the chiral symmetry behavior in finite lattice spacing non-negligibly. So these previous conclusions of  $U(1)_A$  restoration need to be careful re-examined
- The previous work in the 2-flavor case by JLQCD [Phys. Rev. D **111**, 114506 (2025)] using the Möbius Domain-Wall fermions is consistent with the scenario that both  $SU(2)_L \times SU(2)_R$  chiral symmetry and  $U(1)_A$  symmetry are effectively restored near  $T_c \approx 165(3)$  MeV.

## Main part of this work

- We extend this investigation to a more realistic  $N_f = 2 + 1$  QCD case adding strange quark
- We improve it to finer lattice size up to  $L48^3 \times T16$
- We measured spatial two point correlation function data and analysed the symmetry restoration behavior for physical point with different beta and chiral limit case by a fixed beta case with varying quark masses and lattice size
- We examine the symmetry breaking by mesons screening masses as a more phenomenologically effective observable followed the idea of work in the 2-flavor case by JLQCD [Phys. Rev. D **111**, 114506 (2025)]

# Symmetry of QCD

QCD:  $\mathcal{Z} = \int \mathcal{D}A \mathcal{D}\bar{q} \mathcal{D}q e^{-S}$ , Gauge Boson+Fermion

$$S = \int d^4x \left[ -\frac{1}{4} \text{Tr} F \wedge \star F + \bar{q}_a (i \not{D} - m_a) q_a \right], \quad a \text{ in } [1, N_f]$$

Specifically, in the up-down quark mass zero limit:

Flavored chiral symmetry  $SU(2)_L \times SU(2)_R$  :

$$q \rightarrow \exp(i\alpha_a T^a + i\beta_a T^a \gamma_5) q, \quad \bar{q} \rightarrow \bar{q} \exp(-i\alpha_a T^a + i\beta_a T^a \gamma_5),$$

$U(1)_A$  Symmetry :

$$q \rightarrow \exp(i\epsilon \gamma_5) q, \quad \bar{q} \rightarrow \bar{q} \exp(i\epsilon \gamma_5),$$

We use the mass difference of the screening masses of different channels to estimate the degree of the symmetry breaking.

# Meson Correlation Functions

We consider the following set of meson operators

$$O_{\Gamma}(x) = \bar{q}(x)\Gamma F^a q(x),$$

$$\Gamma \in \left\{ \mathbf{1}(S), \gamma_5(P), \gamma_{\mu}(V), \gamma_{\mu}\gamma_5(A), \sigma_{\mu\nu} = \frac{i}{2}[\gamma_{\mu}, \gamma_{\nu}](T), \sigma_{\mu\nu}\gamma_5(X) \right\}$$

And we measure the two-points correlation function of the meson operators of different channels. According to the symmetry properties, the asymptotic behavior of the 2-point correlation function at large distances should be approximately be  $a\text{Cosh}(m_{\Gamma}(x - L/2))$

# Symmetry of Correlation Functions

- Flavored chiral symmetry  $SU(2)_L \times SU(2)_R$  :  $\delta_C O_\Gamma^a = \frac{i}{2} \beta_b \bar{q} \left( \{\Gamma, \gamma_5\} \frac{1}{2} \delta^{ab} \mathbf{1} + [\Gamma, \gamma_5] i \epsilon^{bac} \frac{\tau^c}{2} \right) q$
- $U(1)_A$  Symmetry :  $\delta_A O_\Gamma^a = i \epsilon \bar{q} \{\Gamma, \gamma_5\} \frac{\tau^a}{2} q$

| $\Gamma$                      | $[\Gamma, \gamma_5]$  | $\{\Gamma, \gamma_5\}$     |
|-------------------------------|-----------------------|----------------------------|
| $\mathbf{1}(S)$               |                       | $\gamma_5$                 |
| $\gamma_5(P)$                 |                       | $\mathbf{1}$               |
| $\gamma_\mu(V)$               | $\gamma_\mu \gamma_5$ |                            |
| $\gamma_\mu \gamma_5(A)$      | $\gamma_\mu$          |                            |
| $\sigma_{\mu\nu}(T)$          |                       | $\sigma_{\mu\nu} \gamma_5$ |
| $\sigma_{\mu\nu} \gamma_5(X)$ |                       | $\sigma_{\mu\nu}$          |

| $\Gamma$                               | Reference Name | Abbr. | Symmetry Correspondences                                              |
|----------------------------------------|----------------|-------|-----------------------------------------------------------------------|
| $\mathbb{I}$                           | Scalar         | S     | $\left. \begin{array}{l} U(1)_A \end{array} \right\}$                 |
| $\gamma_5$                             | Pseudo Scalar  | PS    |                                                                       |
| $\gamma_1, \gamma_2$                   | Vector         | V     | $\left. \begin{array}{l} SU(2)_L \times SU(2)_R \end{array} \right\}$ |
| $\gamma_1 \gamma_5, \gamma_2 \gamma_5$ | Axial Vector   | A     |                                                                       |
| $\gamma_4 \gamma_3$                    | Tensor         | $T_t$ | $\left. \begin{array}{l} U(1)_A \end{array} \right\}$                 |
| $\gamma_4 \gamma_3 \gamma_5$           | Axial Tensor   | $X_t$ |                                                                       |
|                                        |                |       | $SU(2)_{CS}$                                                          |



# Order of Chiral Phase Transition

- Perturbative  $4 - \epsilon$  expansion Renormalization Group Analysis of effective linear sigma model for  $\langle \bar{q} P_L q \rangle$ . They found for  $N_f > \sqrt{3}$  flavor symmetry, there's no stable IR fixed point. [Pisarski-Wilczek (1983)]
- The anomaly effect term is a determinant term in the effective linear sigma model, which is a  $N_f$ -th order term. For  $N_f \geq 3$  flavor case, it doesn't change the order of the transition.
- But for  $N_f = 2$  flavor case, it becomes a quadratic term, which can smooth out the singularity and change the transition to second order. If the anomaly effect still exists, the transition can be  $O(4)$  class. If the anomaly effect is suppressed and  $U(1)_A$  symmetry is restored before or at the critical temperature  $T_c$ , then transition should be still first order
- But recently higher-loop computations and FRG computations suggested different possibilities [E.Vicari ,F.Basile, A.Pelissetto 2005 , G. Fejos 2022], and Conformal Bootstrap results indicate some interesting points too[Nakayama Ohtsuki 2016]

# Possible Emergent Larger Symmetry

- Perturbative analysis of screening masses from NRQCD<sub>3</sub> [Laine, M., Vepsäläinen, M. (2004)]:  
 $\left[2M - \frac{\nabla^2}{\pi T} + V(\mathbf{r})\right] \Psi_0 = m_{\text{scr}} \Psi_0$ , where  $M$  is the mass parameter in model
- $M = \pi T + g^2 T \frac{C_F}{8\pi} + \mathcal{O}(g^4 T)$
- We write  $(m_{\text{src}} - 2M)$  as  $g_E^2 \frac{C_F}{2\pi} \hat{E}_0$ , where  $\hat{E}_0$  (around 0.38237 for 2-flavor) is computed numerically
- For two flavor case, we got  $m_{\text{scr}} = 2\pi T + g_E^2 \frac{C_F}{2\pi} (1/2 + \hat{E}_0) = 2\pi T + 0.1872 g^2 T$
- It predicts the flavor non-singlet channel will tend to behave same screening mass. Thus, a larger symmetry, Chiral-Spin Symmetry  $SU(2)_{CS}$ , may emerge

$$q \rightarrow \exp(i\Sigma^i \epsilon_i) q,$$

$$\bar{q} \rightarrow \bar{q} \gamma_0 \exp(-i\Sigma^i \epsilon_i) \gamma_0,$$

where  $\Sigma^i$  has three components,

$$\Sigma^i = (\gamma_k, -i\gamma_5 \gamma_k, \gamma_5), \quad k = 1, 2.$$

# Lattice QCD Numerical Setup

- Möbius Domain-Wall Fermions: The effective 4D operator is [Brower, R. C., Neff, H., & Orginos, K. (2012)]:

$$D_{\text{DW}}^{4\text{D}}(m) = \frac{1+m}{2} + \frac{1-m}{2} \gamma_5 \tanh(L_s \tanh^{-1}(H_M)),$$

where  $m$  is the bare quark mass, and the Möbius kernel  $H_M$  is  $H_M = \gamma_5 \frac{2D_W}{2+D_W}$

The precision of chiral symmetry could be estimated by Residual Mass  $m_{res}$ , and at low beta case is not so good:

| $T$ (MeV)     | 138     | 145     | 153     | 157     | 164.5   | 176     | 202     | 241.6   |
|---------------|---------|---------|---------|---------|---------|---------|---------|---------|
| $m_{res}/m_l$ | 0.9087  | 0.5360  | 0.3394  | 0.2744  | 0.1836  | 0.1041  | 0.0299  | 0.0049  |
| $m_{res}$     | 7.31e-4 | 4.98e-4 | 3.40e-4 | 2.80e-4 | 1.91e-4 | 1.08e-4 | 2.81e-5 | 3.75e-6 |

# Lattice QCD Numerical Setup at physical point

The simulation is run with lattice size  $48^3 \times 16$  (near the physical point):

| $\beta$ | $T$ (MeV) | $am_s$   | $am$     |
|---------|-----------|----------|----------|
| 4.13    | 138       | 0.043547 | 0.000805 |
| 4.15    | 145       | 0.040843 | 0.000930 |
| 4.17    | 153       | 0.038400 | 0.001001 |
| 4.18    | 157       | 0.037265 | 0.001022 |
| 4.20    | 164.5     | 0.035150 | 0.001041 |
| 4.23    | 176       | 0.032315 | 0.001033 |
| 4.30    | 202       | 0.026930 | 0.000939 |

# Lattice Setup at Fixed $\beta = 4.17$

To study the symmetry restoration mechanism in the chiral limit, we perform simulations at a fixed coupling  $\beta = 4.17$ , varying the bare light quark mass  $am_l$  and the temporal lattice size  $N_t$ :

| Lattice Size     | $am_s$ | $am_l$                         | $T$ (MeV) |
|------------------|--------|--------------------------------|-----------|
| $32^3 \times 12$ | 0.040  | 0.0020, 0.0035, 0.0070, 0.0120 | 220.00    |
| $32^3 \times 14$ | 0.040  | 0.0020, 0.0035, 0.0070, 0.0120 | 188.57    |
| $32^3 \times 16$ | 0.040  | 0.0020, 0.0035, 0.0070, 0.0120 | 165.00    |
| $36^3 \times 18$ | 0.040  | 0.0020, 0.0035, 0.0070, 0.0120 | 146.67    |
| $40^3 \times 16$ | 0.040  | 0.0020, 0.0035, 0.0070, 0.0120 | 165.00    |
| $48^3 \times 18$ | 0.040  | 0.0020, 0.0035, 0.0070, 0.0120 | 146.67    |

# Lattice Measurements & Configurations

- We performed measurements for each beta parameters, each channels, and each directions
- To improve the quality of data, we performed **16-source** measurements
- The total number of configurations ( $N_{\text{conf}}$ ) at physical point analyzed for each temperature/ $\beta$  are counted below :

| $\beta$                                     | 4.13 | 4.15 | 4.17 | 4.18 | 4.20  | 4.23 | 4.30 |
|---------------------------------------------|------|------|------|------|-------|------|------|
| $T$ (MeV)                                   | 138  | 145  | 153  | 157  | 164.5 | 176  | 202  |
| No. of Confs ( $N_{\text{conf}}$ )          | 395  | 398  | 396  | 402  | 407   | 395  | 404  |
| Total Meas. ( $16 \times N_{\text{conf}}$ ) | 6320 | 6368 | 6336 | 6432 | 6512  | 6320 | 6464 |

# Meson Channels & Directional Averaging

- To improve the data quality, the correlation functions of each meson channel are averaged over all equivalent spatial directions and components.
- The directional averaging components is summarized below:

| Channel                   | Components | Averaged Directions (configured in <code>config.py</code> )                  |
|---------------------------|------------|------------------------------------------------------------------------------|
| <b>AV</b> (Axial Vector)  | 6          | $X$ (AVector2, AVector3), $Y$ (AVector1, AVector3), $Z$ (AVector1, AVector2) |
| <b>Vec</b> (Vector)       | 6          | $X$ (Vector2, Vector3), $Y$ (Vector1, Vector3), $Z$ (Vector1, Vector2)       |
| <b>S</b> (Scalar)         | 3          | $X, Y, Z$ (TVector4)                                                         |
| <b>PS</b> (Pseudo-Scalar) | 3          | $X, Y, Z$ (TAVector4)                                                        |
| <b>Tt</b> (Tensor)        | 3          | $X$ (TVector1), $Y$ (TVector2), $Z$ (TVector3)                               |
| <b>Xt</b> (Axial Tensor)  | 3          | $X$ (TAVector1), $Y$ (TAVector2), $Z$ (TAVector3)                            |

# Extracting Screening Masses: Methodology

- **Local Effective Mass:** Numerically solve the ratio at each spatial coordinate  $x$  to obtain the local effective mass  $m_{\text{eff}}(x)$ :

$$\frac{C(x)}{C(x+1)} = \frac{\cosh(m_{\text{eff}}(x)(x - L/2))}{\cosh(m_{\text{eff}}(x)(x+1 - L/2))}$$

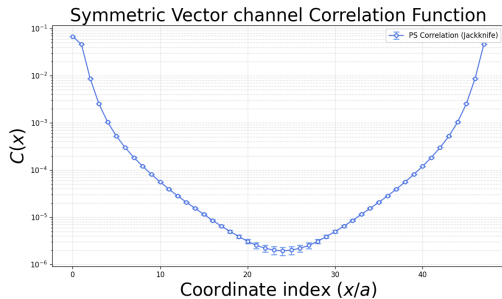
- **Plateau and Range:** The plateau of  $m_{\text{eff}}(x)$  at large separations is used to determine a suitable fitting range  $[x_{\text{min}}, x_{\text{max}}]$ .
- **Screening Mass Fit:** The physical screening mass  $m_{\text{scr}}$  is extracted by performing a correlated  $\chi^2$  fit of  $C(x)$  to the asymptotic form:

$$C(x) = A \cosh \left( m_{\text{scr}} \left( x - \frac{L}{2} \right) \right)$$

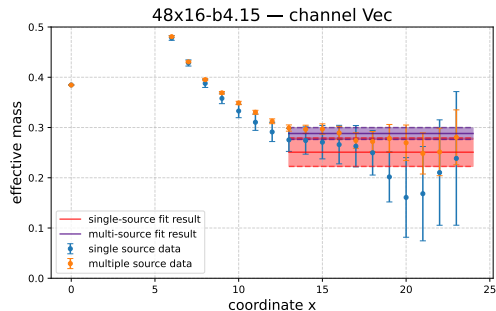
- **Statistical Error:** Jackknife method and Binning analysis are used to handle statistical uncertainties and eliminate HMC autocorrelation bias.



# Extracting Screening Masses: Fit Examples

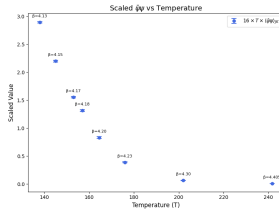
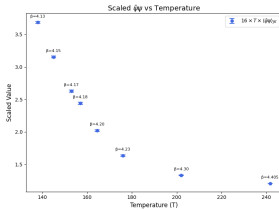


correlation data of Vec channel for  
 $\beta = 4.15(T=153\text{MeV})$



effective mass of Vec channel for  
 $\beta = 4.15(T=153\text{MeV})$

# Chiral Condensate & Crossover Phase Transition



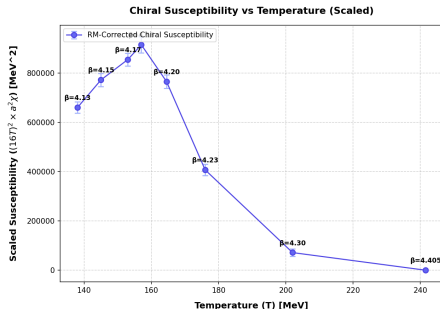
Left: Bare Chiral Condensate, Right: Renormalized Chiral Condensate

## Definition:

$$a^3 \langle \bar{\psi} \psi_q \rangle = \frac{1}{N_s^3 N_\tau} \frac{N_f}{4} \langle \text{Tr} \not{D}_q^{-1} \rangle$$

- Order parameter for spontaneous chiral symmetry breaking
- We subtract strange quark part to remove divergence

# Disconnected Chiral Susceptibility



Disconnected susceptibility  $\chi_{\text{dis}}$  vs Temperature

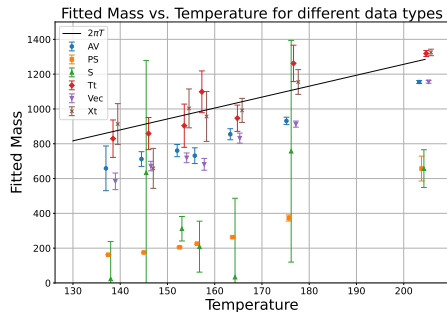
Definition:

$$a^2 \chi_{\text{dis}} = \frac{1}{N_s^3 N_\tau} \left( \frac{N_f}{4} \right)^2 \left( \langle (\text{Tr} \not{D}^{-1})^2 \rangle - \langle \text{Tr} \not{D}^{-1} \rangle^2 \right)$$

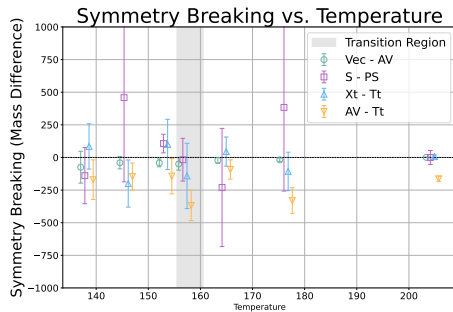
- Fluctuations of order parameter, related to topological effects.
- Clear peak determines pseudo-critical temperature:  $T_c \approx 158 \text{ MeV}$ .

# Numerical Results

Multi-source analysis results (grey band shows 2-flavor  $T_c = 158(3)$  MeV range, while for chiral limit I draw  $T_c \approx 165(3)$  MeV [PRD 111, 114506 (2025)]):



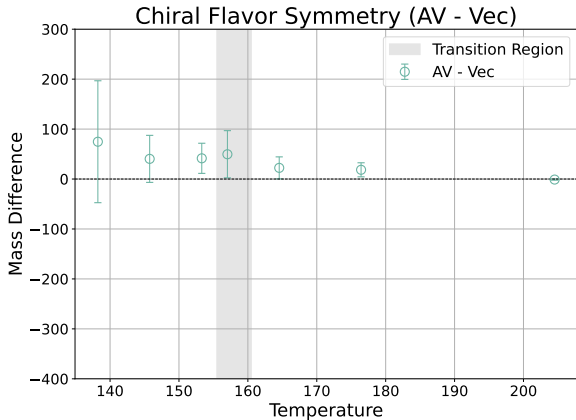
simulated effective masses of different channels



Symmetry breaking of different channels

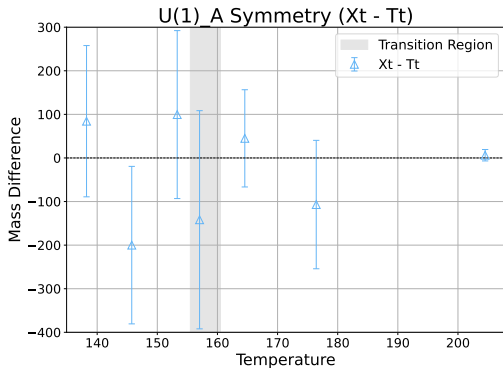
## $SU(2)_L \times SU(2)_R$ Chiral Symmetry

- This indicates the restoration of the  $SU(2)_L \times SU(2)_R$  chiral symmetry (e.g., AV vs Vec channels).
- At  $T = 0$ , the difference between  $a_1$  (1230 MeV) and  $\rho$  meson (775 MeV) is around 455 MeV.
- But the pseudo critical temperature is not very clear (due to finite quark mass)

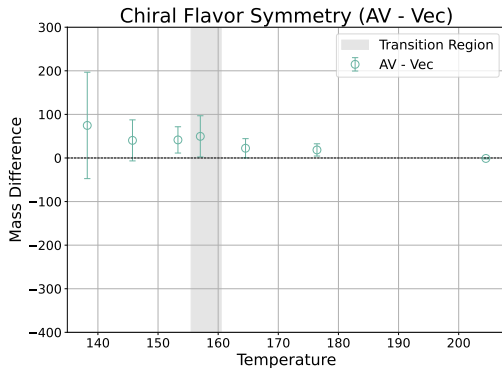


## $U(1)_A$ Symmetry ( $X_t - T_t$ )

The difference tends to decrease at high temperatures, suggesting approximate restoration of  $U(1)_A$  symmetry (anomaly suppression), compared to the flavour symmetry :

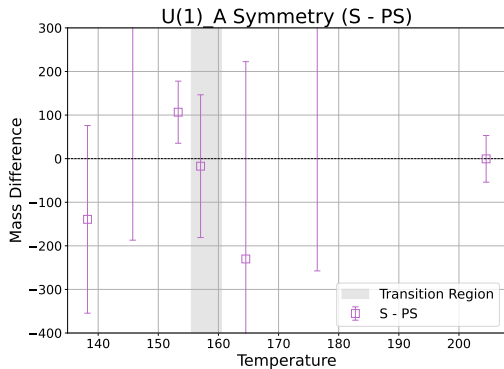


Axial Tensor vs Tensor ( $U(1)_A$ )

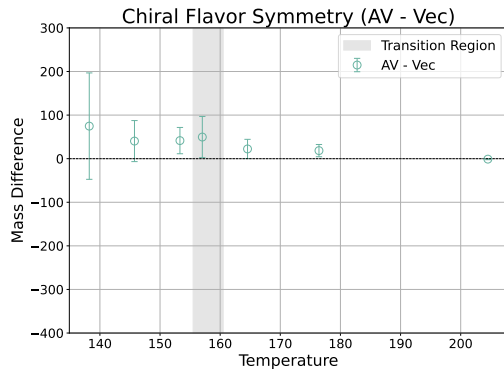


Vector vs Axial-Vector (Flavor Symmetry)

# $U(1)_A$ Symmetry ( $S - PS$ )



Scalar vs Pseudo-Scalar ( $U(1)_A$ )

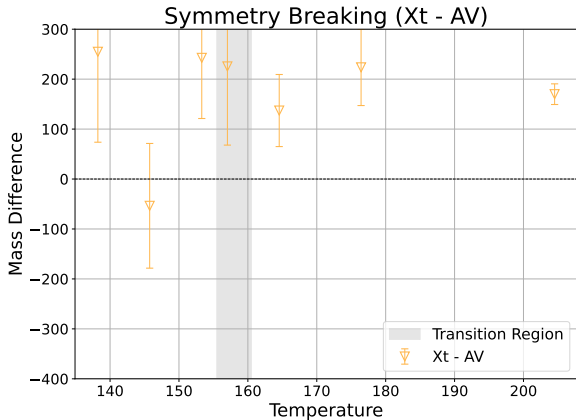


Vector vs Axial-Vector (Flavor Symmetry)

## $SU(2)_{CS}$ Symmetry

Observation: Chiral-Spin (CS) symmetry is not yet completely visible at the temperatures analyzed near the phase transition.

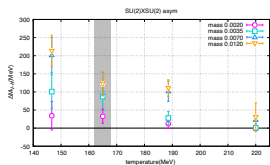
Requires exploration of higher temperature domains.



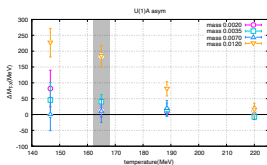


# Symmetry Restoration at Fixed $\beta = 4.17$

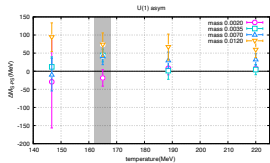
Asymmetry observables as a function of temperature for various light quark masses:



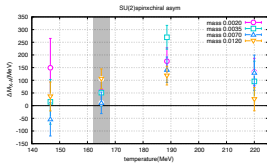
$SU(2)_L \times SU(2)_R$  (V-AV)



$U(1)_A$  (T-X)



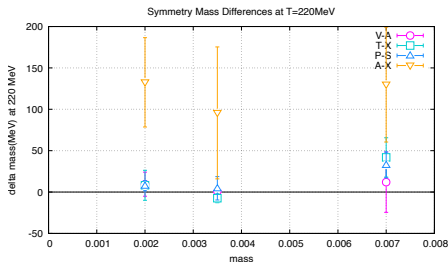
$U(1)$  (S-PS)



$SU(2)_{CS}$  (X-AV)

- All symmetries show a clear trend towards restoration above  $T_c$ .
- $U(1)_A$  and chiral  $SU(2)_L \times SU(2)_R$  symmetries show an almost simultaneous restoration as we approach the chiral limit.
- Chiral-spin symmetry  $SU(2)_{CS}$  remains broken in the analyzed temperature range.

# Chiral Limit Extrapolation ( $T \approx 220$ MeV)



Mass differences between mesonic partners as a function of  $am_l$  at  $T \approx 220$  MeV

- Analyzed mass differences at a high-temperature snapshot ( $T \approx 220$  MeV) as a function of the bare light quark mass  $am_l$ .
- As  $am_l \rightarrow 0$ , the mass differences between chiral partners ( $m_{\text{Vec}} - m_{\text{AV}}$  and  $m_{\text{S}} - m_{\text{PS}}$ ) extrapolate toward zero.
- This supports the scenario of a **simultaneous restoration** of chiral flavor and axial  $U(1)_A$  symmetries in the chiral limit.

# Summary and Outlook

- **Physical Point:** Crossover transition with  $T_c \approx 158$  MeV. Above  $T_c$ , chiral and  $U(1)_A$  symmetry breakings are consistent with zero within errors.
- **Chiral Limit Study:** Extrapolations to  $am_l \rightarrow 0$  support simultaneous restoration of chiral flavor and  $U(1)_A$  symmetries (with  $T_c \approx 165$  MeV).
- **Chiral-Spin Symmetry:** Remains broken up to 220 MeV, indicating active non-perturbative chromomagnetic interactions in QGP.
- **Transition Scenario:**
  - **Previous:** 1st-order if  $U(1)_A$  restored; 2nd-order  $O(4)$  if  $U(1)_A$  remains broken.
  - **New:**  $O(4)$  if  $U(1)_A$  remains; weak 1st-order, or 2nd-order ( $O(4) \times Z_2$ ,  $O(4) \times Z_4$ ,  $O(4) \times O(2)$ ) if  $U(1)_A$  restored.

The symmetry restoration behavior of  $U(1)_A$  and flavor  $SU(2)_L \times SU(2)_R$  will determine the final universality class